

Lesson Objective

Decompose fractions using area models to show equivalence

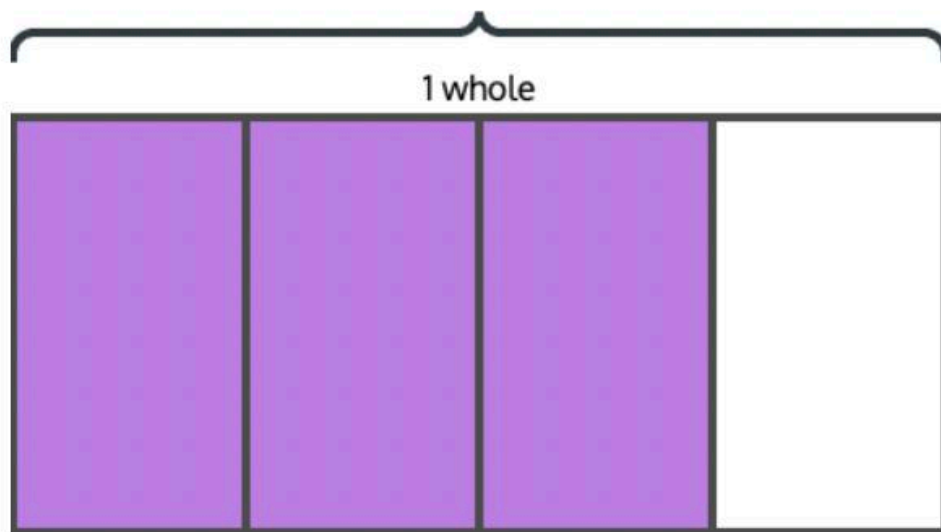
Materials

(S) Personal white board, Mini-Lesson Worksheet

PROBLEM

1 Use an area model to decompose fourths into eighths, and write addition and multiplication sentences to show the equivalence.

The rectangle below represents 1 whole and is shaded to show $\frac{3}{4}$.



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Draw a horizontal line to decompose the fourths into eighths. Then, write an addition sentence using parentheses and a multiplication sentence to show that $\frac{3}{4} = \frac{6}{8}$.

Answer: $(1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8$; $6 \times 1/8 = 6/8$;
 $3/4 = 6/8$

TEACHER GUIDANCE

Tell students that the rectangle represents 1 whole with $\frac{3}{4}$ shaded.

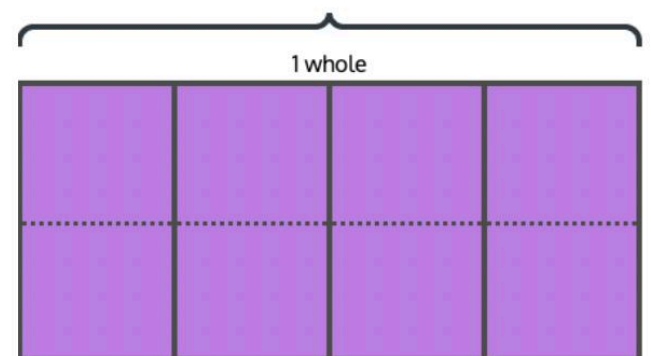
Ask students how many fourths are shaded and how they could split each fourth into 2 equal parts.

Model drawing one horizontal line across the rectangle to create eighths. Ask students how many eighths are shaded now.

Guide students through writing an addition sentence, using parentheses to group the 2 eighths that make up each fourth: $(1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8$.

Guide students through writing a multiplication sentence: $6 \times \frac{1}{8} = \frac{6}{8}$.

Point out that the shaded area did not change — the number of pieces increased and each piece got smaller, but the total amount stayed the same, so $\frac{3}{4} = \frac{6}{8}$.



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PROBLEM

- 2 **Draw an area model to show that two-thirds is equivalent to eight-twelfths, and express the equivalence using addition and multiplication of unit fractions.**

Draw an area model to show that $\frac{2}{3} = \frac{8}{12}$. Write an addition sentence using parentheses and a multiplication sentence to prove the equivalence.

Find the slope of line a .

Answer: $(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) + (\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) = \frac{8}{12}$; $8 \times \frac{1}{12} = \frac{8}{12}$; $\frac{2}{3} = \frac{8}{12}$

TEACHER GUIDANCE

Ask students which fraction to draw first and why. Have students draw a rectangle representing 1 whole, partition it into thirds, and shade $\frac{2}{3}$.

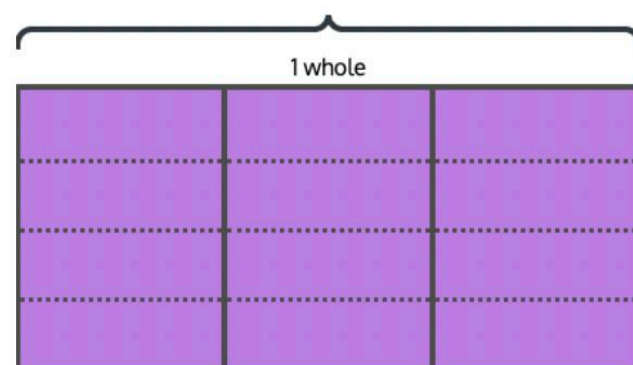
Ask students how they can decompose thirds into twelfths. Guide students to see that each third must be split into 4 equal parts, so they draw 3 horizontal lines across the model.

Ask students how many total parts are in the whole now and how many are shaded.

Have students write an addition sentence using parentheses to group the 4 twelfths within each third: $(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) + (\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}) = \frac{8}{12}$.

Have students write a multiplication sentence: $8 \times \frac{1}{12} = \frac{8}{12}$.

Ask students what these sentences tell them about $\frac{2}{3}$ and $\frac{8}{12}$. Confirm that the shaded area is the same — $\frac{2}{3}$ and $\frac{8}{12}$ are equivalent fractions.



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- 3 **Draw an area model to represent a fraction greater than one, then partition it into smaller units to find an equivalent fraction.**

Draw an area model to represent $\frac{5}{3}$. Then, partition the thirds into sixths to find an equivalent fraction.

Answer: $\frac{5}{3} = \frac{10}{6}$

Students work independently.

Monitor For

- Do students draw the horizontal line(s) across the entire rectangle, creating equal-sized parts in every column, not just in the shaded region?
- Do students correctly count the total number of smaller parts in the whole to determine the new denominator (e.g., 8 parts total means eighths, not just counting shaded parts)?
- Do students use parentheses to group the unit fractions that compose each original unit (e.g., grouping the 2 eighths within each fourth)?
- In Problem 3, do students extend the area model beyond 1 whole to represent $5/3$, drawing a second rectangle for the remaining 2 thirds?

Instructional Moves

- If a student only draws lines in the shaded region, ask how many equal parts must be in the whole and have them extend the lines across the entire rectangle.
- Before students write parentheses, have them circle the smaller unit fractions on the area model that came from one original unit — this connects the grouping to what they see.
- Remind students that $5/3$ means 5 copies of $1/3$ — draw 1 whole (3 thirds) and then 2 more thirds in a second rectangle to represent all 5 thirds.
- If a student writes $6 \times 1/8$ but cannot explain why, ask them to count the shaded eighths one by one and connect each count to the addition sentence.

Reflection Questions

1. When we drew a horizontal line to split each fourth into eighths, what happened to the number of parts? What happened to the size of each part?
Answer: The number of parts doubled — there were more parts, but each part was smaller. The total shaded area stayed the same.
2. Why did we use parentheses when writing the addition sentence for $3/4 = 6/8$?
Answer: The parentheses group the eighths that make up each fourth, so we can see that each $1/4$ is the same as 2 copies of $1/8$.
3. How did you represent $5/3$ with an area model, and how did you know it would need more than 1 whole?
Answer: $5/3$ means 5 copies of $1/3$, but 1 whole only has 3 thirds, so I needed a second rectangle for the extra 2 thirds.